

Операционные системы

Командная оболочка Midnight Commander

Булут Умут

2026-03-25

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Изучить возможности Midnight Commander
- 2 Изучить редактор Midnight Commander

2. Процесс выполнения лабораторной работы

Работа с Midnight Commander

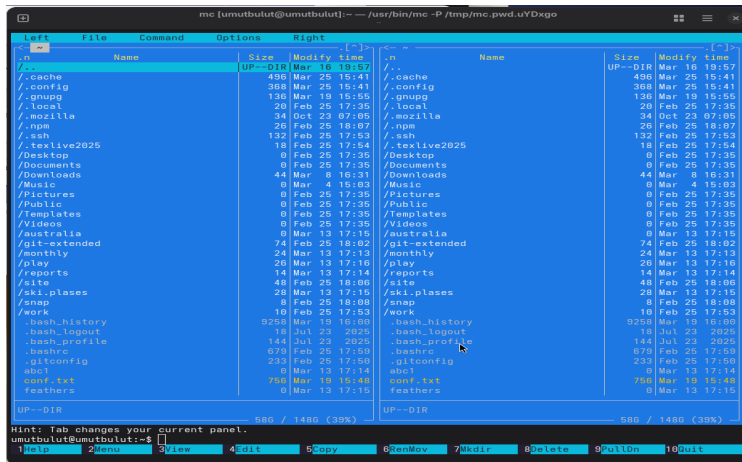


Рисунок 1: Запуск mc

Работа с Midnight Commander

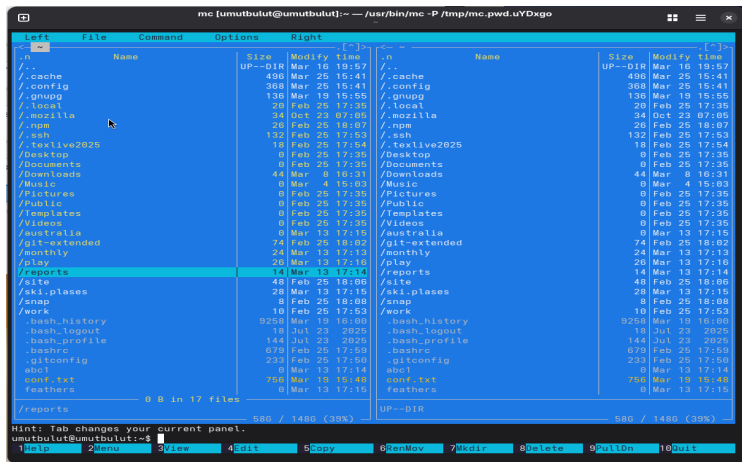


Рисунок 2: Выделение

Работа с Midnight Commander

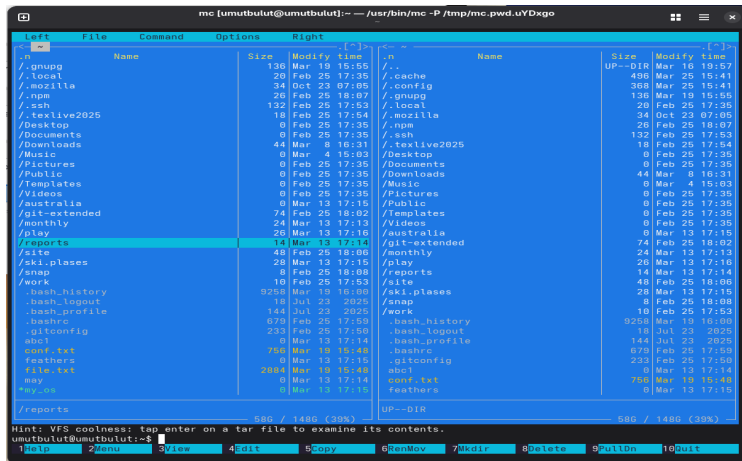


Рисунок 3: Отмена

Работа с Midnight Commander

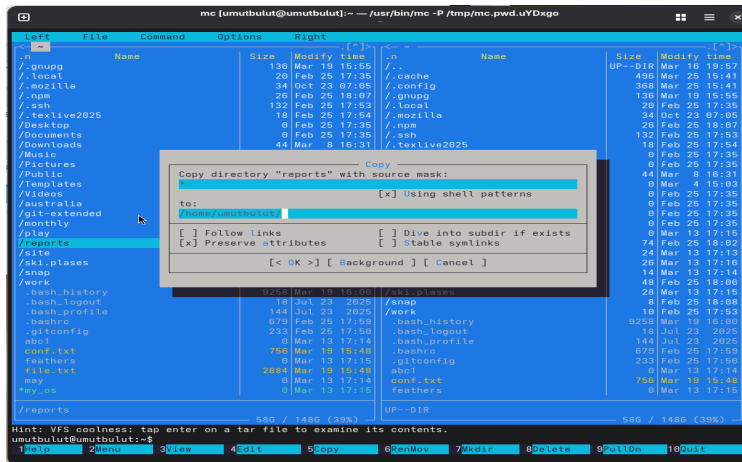


Рисунок 4: Копирование

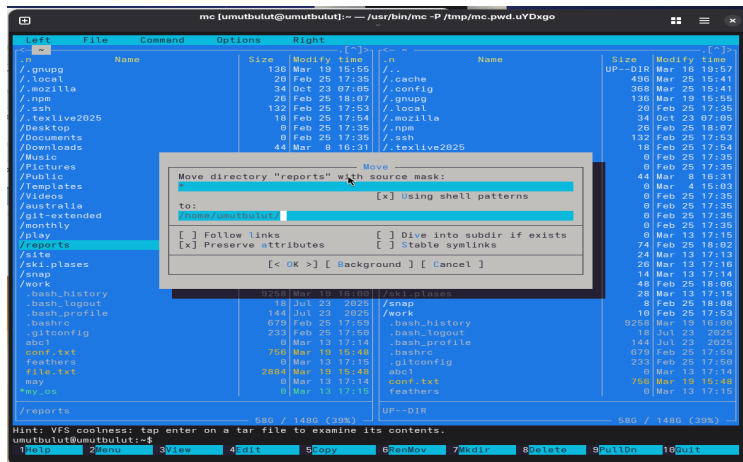


Рисунок 5: Перемещение

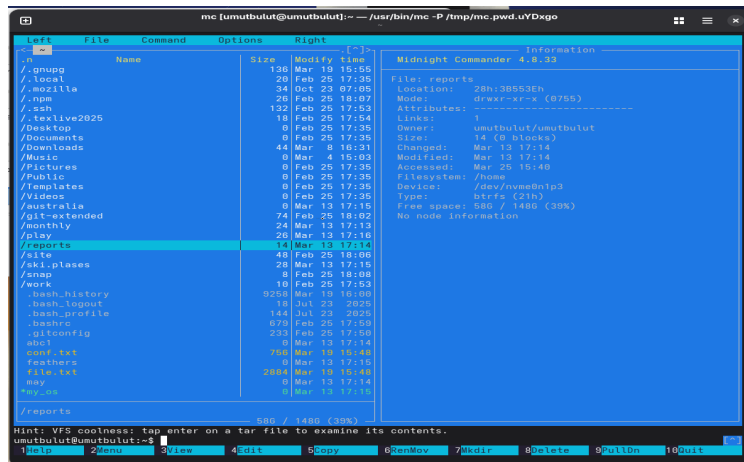


Рисунок 6: Информация

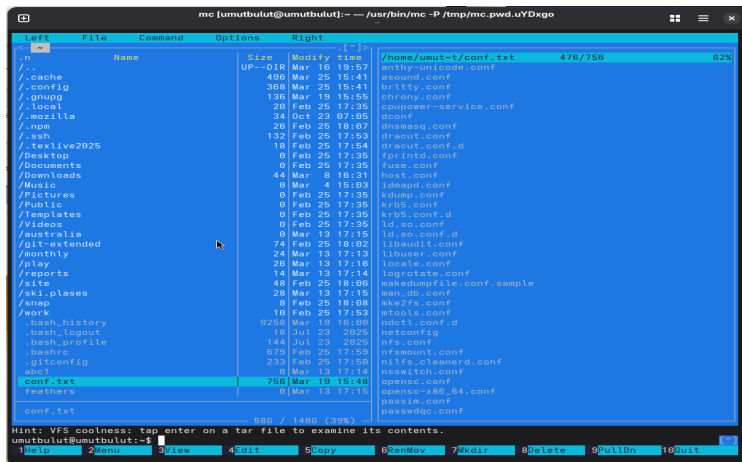


Рисунок 7: Быстрый просмотр

Работа с Midnight Commander

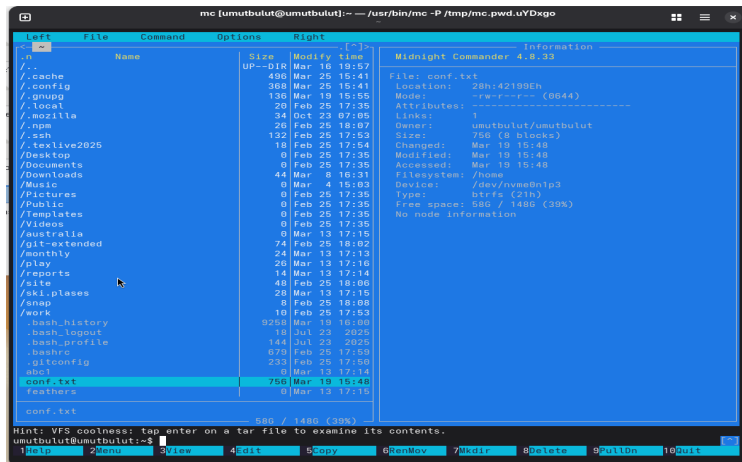


Рисунок 8: Информация

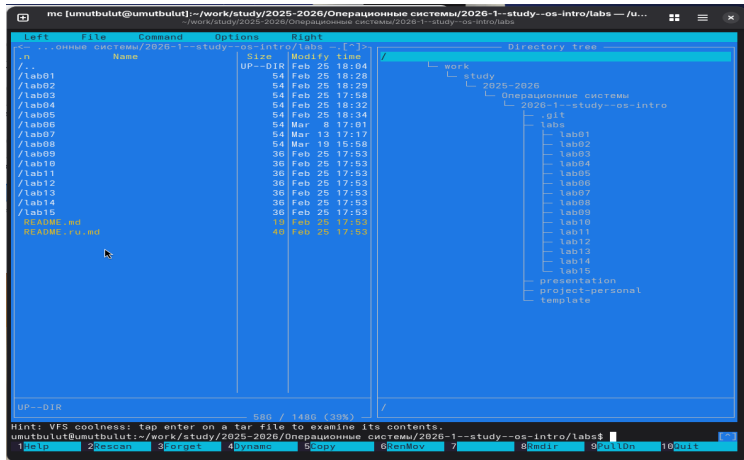


Рисунок 9: Дерево каталогов



The screenshot shows a terminal window of the Midnight Commander (MC) file manager. The title bar indicates the user is 'umutbulut' on a system named 'umutbulut', and the current directory is '/work/study/2025-2026/Операционные системы/2026-1--study--os-intro'. The terminal displays the contents of a file named 'Makefile' located at '/home/umutbulut/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/'. The file content includes a shebang line, course information, a help section, and several targets: 'list', 'prepare', and 'submodule'. The 'list' target lists course files, 'prepare' generates a directory structure, and 'submodule' updates submodules. The bottom status bar shows navigation shortcuts: 1 Help, 2 UnWrap, 3 Quit, 4 Lex, 5 Goto, 6, 7 Search, 8 Raw, 9 Format, 10 Quit.

```
mc [umutbulut@umutbulut]:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro — /usr/bi...
~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro
/home/umutbulut/work/study/2025-2026/Операционные системы/2026-1--study--os-intro/Makefile 1021/1021 100%
SHELL := /bin/bash
COURSE =

.PHONY: all clean

all: help

help:
    @echo 'Usage:'
    @echo ' make <target>'
    @echo
    @echo 'Targets:'
    @grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN
(FS = ".*?###"); {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | cut -d: -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
    @grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = ".*?###"}
); {printf " \033[36m%-30s\033[0m %s\n", $$1, $$2}'
    @echo

list:    ## List of courses
    @. ./template/config/script/list-courses

prepare:    ## Generate directories structure
    @. ./template/config/script/prepare
    @touch prepare

submodule:    ## Update submules
    git submodule update --init --recursive
    git submodule foreach 'git fetch origin; git checkout $$ (git rev-parse --abbrev-ref HEAD); git reset --har
d origin/$(git rev-parse --abbrev-ref HEAD); git submodule update --recursive; git clean -dfx'
```

Рисунок 10: Просмотр содержимого текстового файла

Работа с Midnight Commander



The screenshot shows the Midnight Commander (mc) interface. The title bar indicates the current directory is `~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro`. The main window displays a `Makefile` with the following content:

```
Makefile
SHELL := /bin/bash
COURSE = .

.PHONY: all clean

all: help

help:
-----@echo 'Usage:'
-----@echo ' make <target>'
-----@echo
-----@echo 'Targets:'
-----@grep -E '^[a-zA-Z_0-9-]+:.*?##.*$$' $(MAKEFILE_LIST) | grep -v '###' | sort | cut -d: -f1- | awk 'BEGIN
-----@grep -E '###.*' $(MAKEFILE_LIST) | cut -d' ' -f2- | awk 'BEGIN {FS = "###"}; {printf "%s\n", $$1, $$2}'
-----@grep -E '^[a-zA-Z_0-9-]+:.*?###.*$$' $(MAKEFILE_LIST) | sort | cut -d: -f2- | awk 'BEGIN {FS = "###"};
-----@echo

list:-----## List of courses
-----@. ./template/config/script/list-courses

prepare:-----## Generate directories structure
-----@. ./template/config/script/prepare
-----@touch prepare

submodule:-----## Update submules
-----@git submodule update --init --recursive
-----@git submodule foreach 'git fetch origin; git checkout $(git rev-parse --abbrev-ref HEAD); git reset --hard'
```

At the bottom of the window, there is a toolbar with icons and labels for various actions: 1 Help, 2 Save, 3 Mark, 4 Replac, 5 Copy, 6 Move, 7 Search, 8 Delete, 9 PullDn, 10 Quit.

Рисунок 11: Отредактируем содержимое текстового файла без сохранения результатов

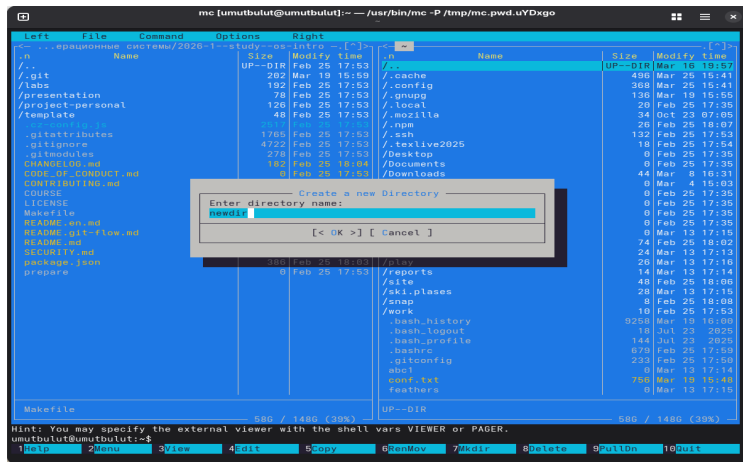


Рисунок 12: Создание каталога

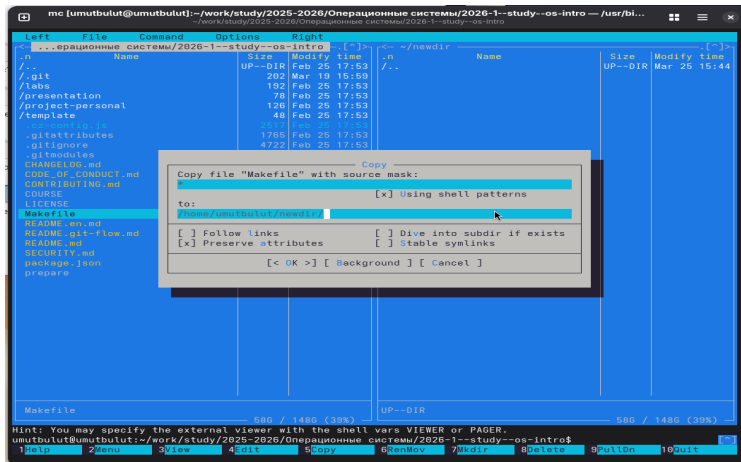


Рисунок 13: Копирование в файлов в созданный каталог

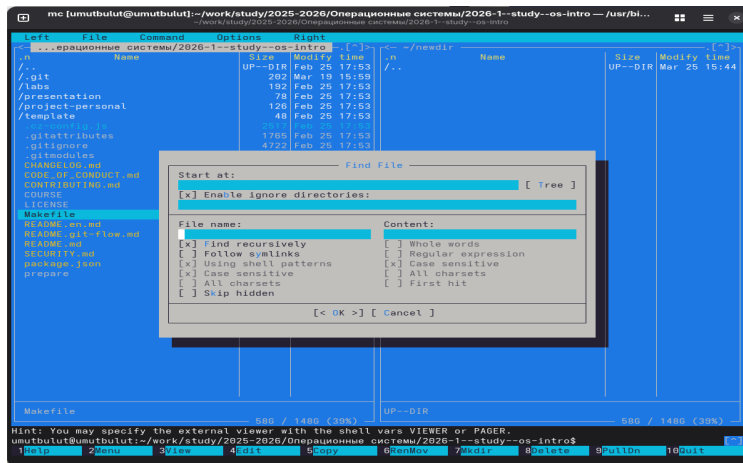


Рисунок 14: Поиск файлов

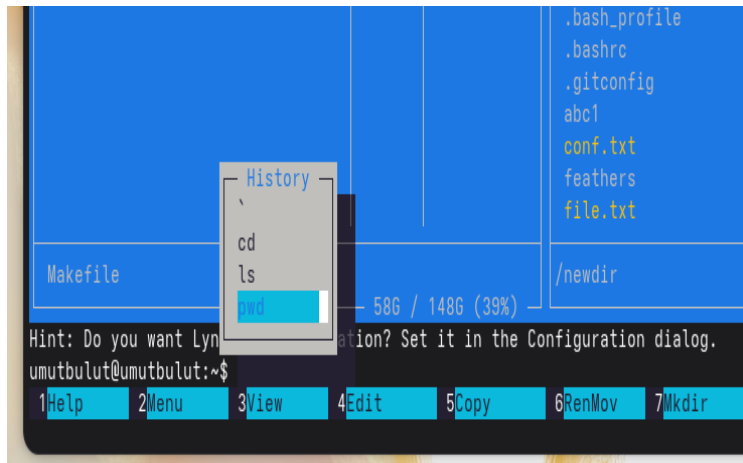


Рисунок 15: История команд

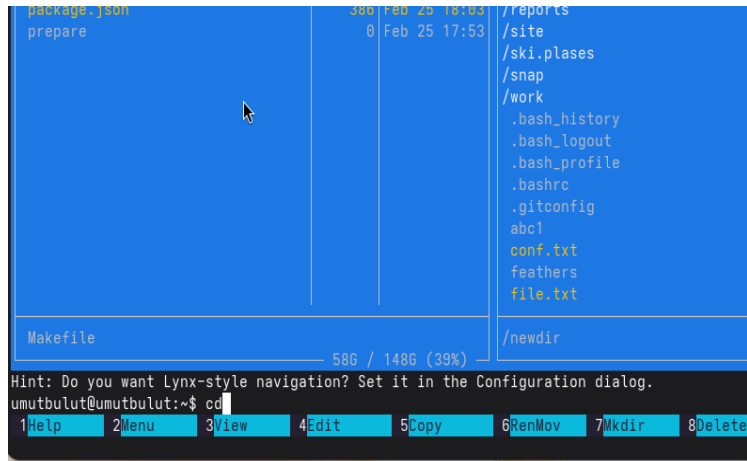
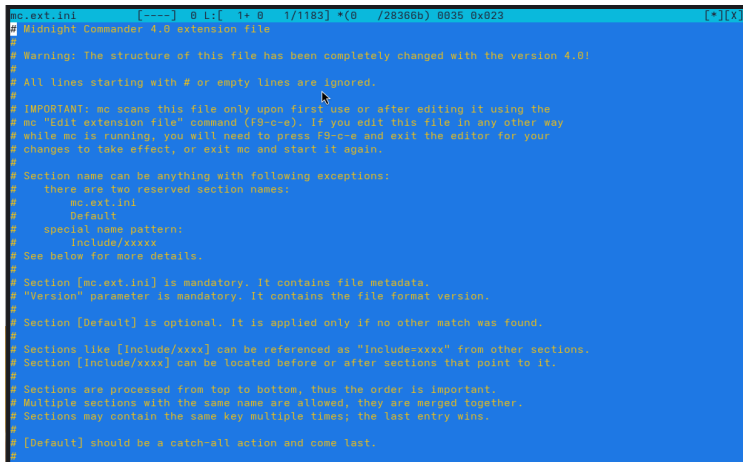


Рисунок 16: Переход в домашний каталог



```
mc.ext.ini      [---] 0 L: 1+ 0 1/1183 *(0 /28366b) 0035 0x023      [*][X]
# Midnight Commander 4.8 extension file
#
# Warning: The structure of this file has been completely changed with the version 4.8!
#
# All lines starting with # or empty lines are ignored.
#
# IMPORTANT: mc scans this file only upon first use or after editing it using the
# mc "Edit extension file" command (F9-c-e). If you edit this file in any other way
# while mc is running, you will need to press F9-c-e and exit the editor for your
# changes to take effect, or exit mc and start it again.
#
# Section name can be anything with following exceptions:
#   there are two reserved section names:
#       mc.ext.ini
#       Default
#   special name pattern:
#       Include/xxxx
# See below for more details.
#
# Section [mc.ext.ini] is mandatory. It contains file metadata.
# "Version" parameter is mandatory. It contains the file format version.
#
# Section [Default] is optional. It is applied only if no other match was found.
#
# Sections like [Include/xxxx] can be referenced as "Include=xxxx" from other sections.
# Section [include/xxxx] can be located before or after sections that point to it.
#
# Sections are processed from top to bottom, thus the order is important.
# Multiple sections with the same name are allowed, they are merged together.
# Sections may contain the same key multiple times; the last entry wins.
#
# [Default] should be a catch-all action and come last.
```

Рисунок 17: Просмотр файла расширений

```
menu [----] 0 L:[ 1+ 0 1/380] *(0 /12204b) 0115 0x073 [X]
shell_patterns=0

#####
# %% The % character
# %f The current file (if non-local vfs, file will be copied locally and
#           %f will be full path to it)
# %p The current file
# %d The current working directory
# %s "Selected files"; the tagged files if any, otherwise the current file
# %t Tagged files
# %u Tagged files (and they are untagged on return from expand_format)
# %view Runs the commands and pipes standard output to the view command
# If %view is immediately followed by '{', recognize keywords
#   ascii, hex, proff and unform
#
# If the format letter is in uppercase, it refers to the other panel
#
# With a number followed the % character you can turn quoting on (default)
# and off. For example:
# %f   quote expanded macro
# %if  ditto
# %0f  don't quote expanded macro
#####

+ ! t t
0   Do something on the current file
    CMD=%{Enter command}
    $CMD %f

+ t t
0   Do something on the tagged files
    CMD=%{Enter command}
    for i in %t ; do
```

Рисунок 18: Просмотр файла меню

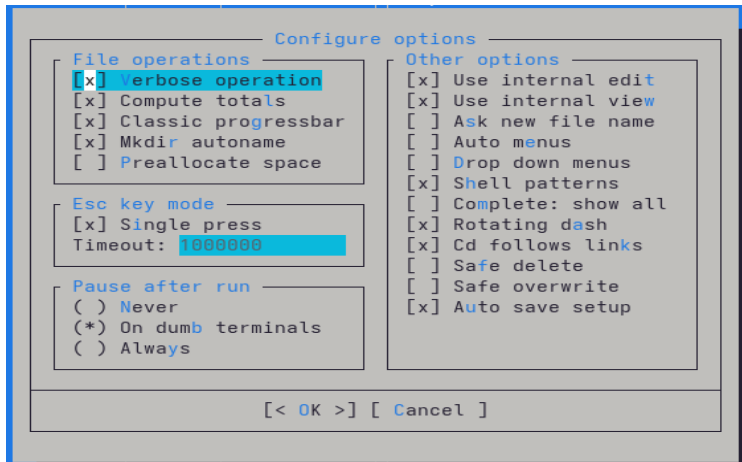


Рисунок 19: Конфигурация

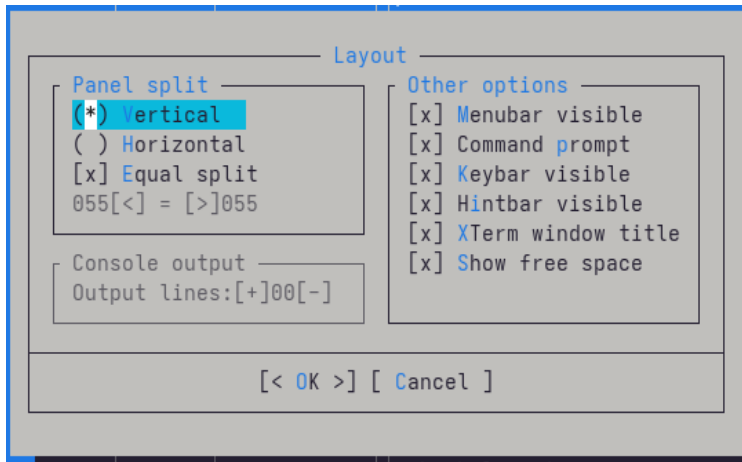


Рисунок 20: Внешний вид

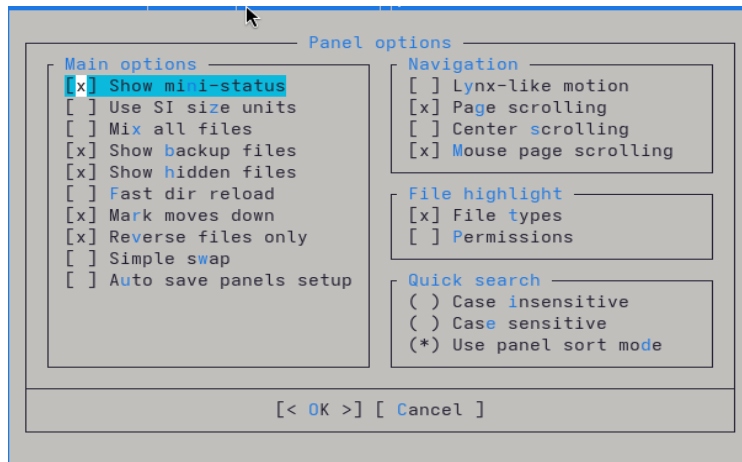


Рисунок 21: Настройки панелей

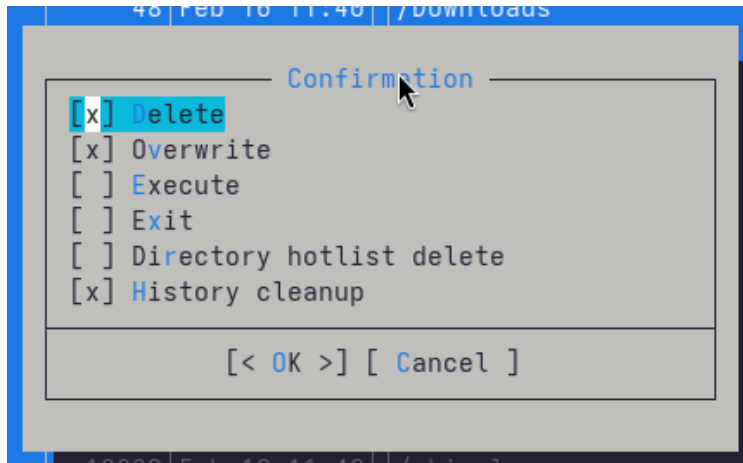


Рисунок 22: Подтверждение

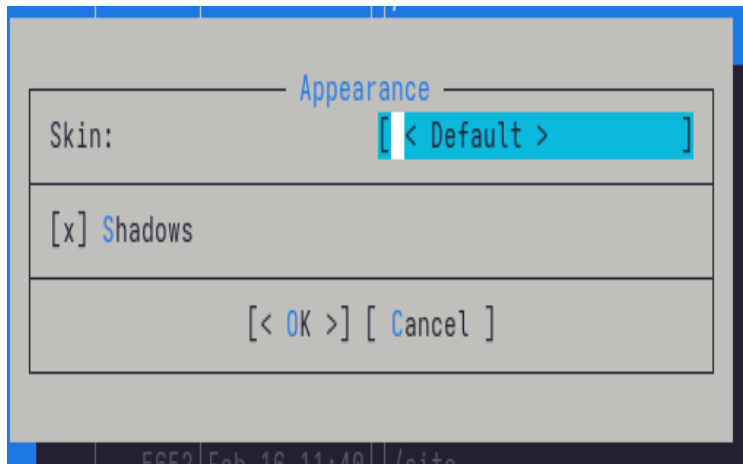


Рисунок 23: Оформление

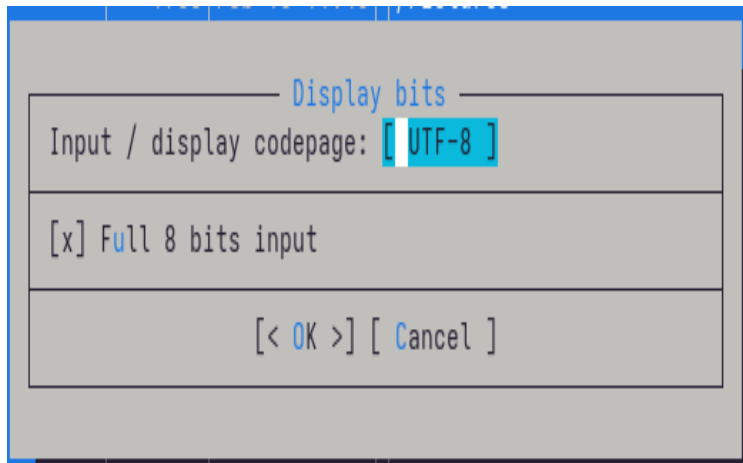


Рисунок 24: Кодировка символов

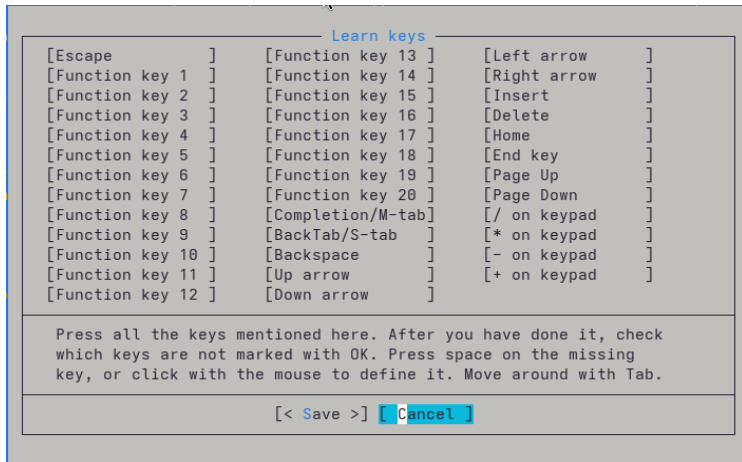
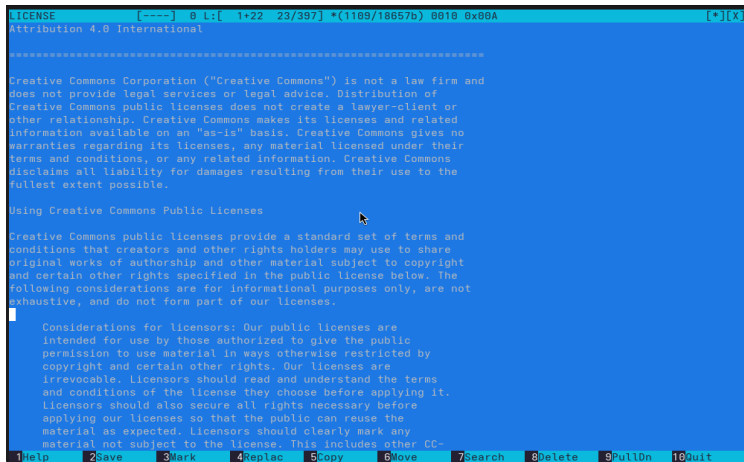


Рисунок 25: Распознавание клавиш



```
LICENSE      [-----]  8 L:[ 1+22 23/397] *(1109/18657b) 0010 0x00A      [*][X]
Attribution 4.0 International

=====

Creative Commons Corporation ("Creative Commons") is not a law firm and
does not provide legal services or legal advice. Distribution of
Creative Commons public licenses does not create a lawyer-client or
other relationship. Creative Commons makes its licenses and related
information available on an "as-is" basis. Creative Commons gives no
warranties regarding its licenses, any material licensed under their
terms and conditions, or any related information. Creative Commons
disclaims all liability for damages resulting from their use to the
fullest extent possible.

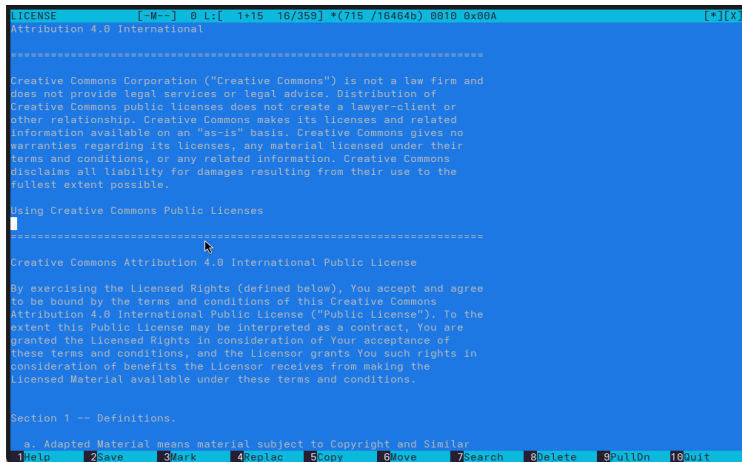
Using Creative Commons Public Licenses

Creative Commons public licenses provide a standard set of terms and
conditions that creators and other rights holders may use to share
original works of authorship and other material subject to copyright
and certain other rights specified in the public license below. The
following considerations are for informational purposes only, are not
exhaustive, and do not form part of our licenses.

  Considerations for licensors: Our public licenses are
  intended for use by those authorized to give the public
  permission to use material in ways otherwise restricted by
  copyright and certain other rights. Our licenses are
  irrevocable. Licensors should read and understand the terms
  and conditions of the license they choose before applying it.
  Licensors should also secure all rights necessary before
  applying our licenses so that the public can reuse the
  material as expected. Licensors should clearly mark any
  material not subject to the license. This includes other CC-

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 26: Файл с текстом



```
LICENSE      [-M--]  8 L:[ 1+15 16/359] *(715 /16464b) 0010 0x00A      [*][x]
Attribution 4.0 International

=====

Creative Commons Corporation ("Creative Commons") is not a law firm and
does not provide legal services or legal advice. Distribution of
Creative Commons public licenses does not create a lawyer-client or
other relationship. Creative Commons makes its licenses and related
information available on an "as-is" basis. Creative Commons gives no
warranties regarding its licenses, any material licensed under their
terms and conditions, or any related information. Creative Commons
disclaims all liability for damages resulting from their use to the
fullest extent possible.

Using Creative Commons Public Licenses
[ ]
=====

Creative Commons Attribution 4.0 International Public License

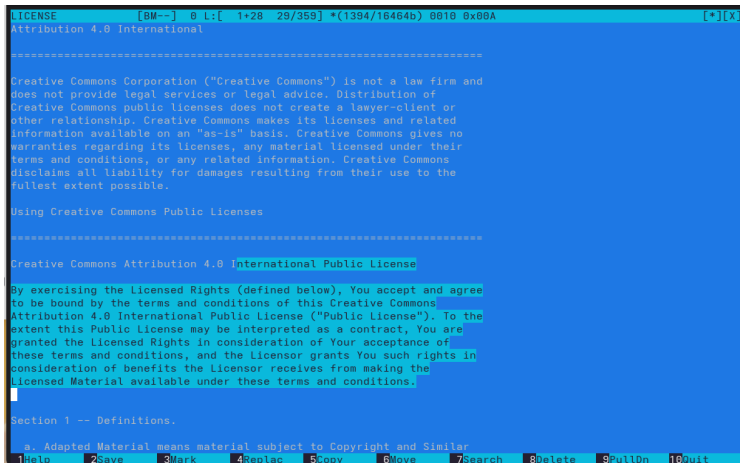
By exercising the Licensed Rights (defined below), You accept and agree
to be bound by the terms and conditions of this Creative Commons
Attribution 4.0 International Public License ("Public License"). To the
extent this Public License may be interpreted as a contract, You are
granted the Licensed Rights in consideration of Your acceptance of
these terms and conditions, and the Licensor grants You such rights in
consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.

Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 27: Файл с текстом



```
LICENSE      [BM--]  0 L:[ 1+28 29/359] *(1394/16464b) 0010 0x00A      [*][X]
Attribution 4.0 International

=====

Creative Commons Corporation ("Creative Commons") is not a law firm and
does not provide legal services or legal advice. Distribution of
Creative Commons public licenses does not create a lawyer-client or
other relationship. Creative Commons makes its licenses and related
information available on an "as-is" basis. Creative Commons gives no
warranties regarding its licenses, any material licensed under their
terms and conditions, or any related information. Creative Commons
disclaims all liability for damages resulting from their use to the
fullest extent possible.

Using Creative Commons Public Licenses

=====

Creative Commons Attribution 4.0 International Public License

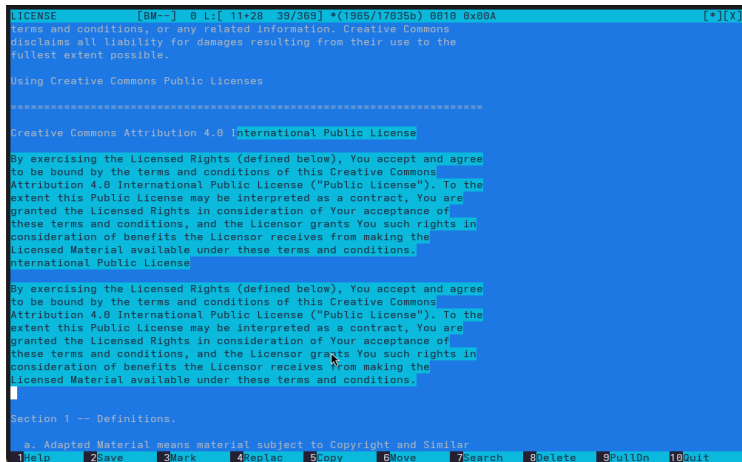
By exercising the Licensed Rights (defined below), You accept and agree
to be bound by the terms and conditions of this Creative Commons
Attribution 4.0 International Public License ("Public License"). To the
extent this Public License may be interpreted as a contract, You are
granted the Licensed Rights in consideration of Your acceptance of
these terms and conditions, and the Licensor grants You such rights in
consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.

Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullOn     10Quit
```

Рисунок 28: Копирование фрагмента



```
LICENSE      [BM--]  0 L:[ 11+28  39/369] *(1965/17035b) 0010 0x00A      [*][X]
terms and conditions, or any related information. Creative Commons
disclaims all liability for damages resulting from their use to the
fullest extent possible.

Using Creative Commons Public Licenses

*****

Creative Commons Attribution 4.0 International Public License

By exercising the Licensed Rights (defined below), You accept and agree
to be bound by the terms and conditions of this Creative Commons
Attribution 4.0 International Public License ("Public License"). To the
extent this Public License may be interpreted as a contract, You are
granted the Licensed Rights in consideration of Your acceptance of
these terms and conditions, and the Licensor grants You such rights in
consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.
International Public License

By exercising the Licensed Rights (defined below), You accept and agree
to be bound by the terms and conditions of this Creative Commons
Attribution 4.0 International Public License ("Public License"). To the
extent this Public License may be interpreted as a contract, You are
granted the Licensed Rights in consideration of Your acceptance of
these terms and conditions, and the Licensor grants You such rights in
consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.

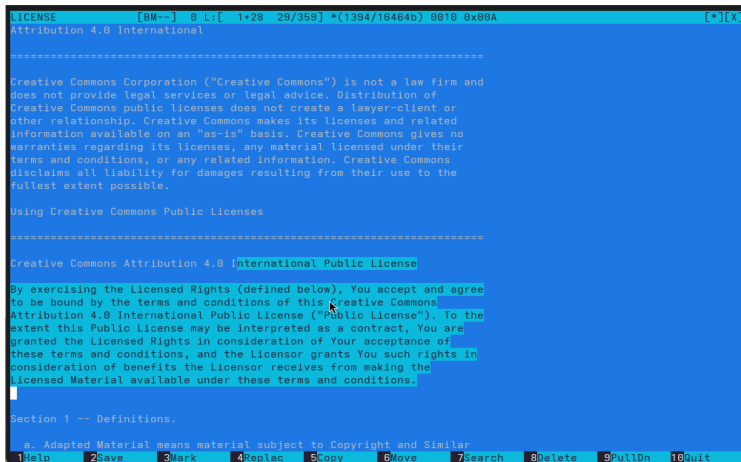
Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help  2Save  3Mark  4Replac  5Copy  6Move  7Search  8Delete  9PullDn 10Quit
```

Рисунок 29: Сохранение

Работа с редактором Midnight Commander



```
LICENSE      [BM--]  8 L:[ 1+28 29/359] *(1394/16464b) 0010 0x00A [~][X]
Attribution 4.0 International

=====

Creative Commons Corporation ("Creative Commons") is not a law firm and
does not provide legal services or legal advice. Distribution of
Creative Commons public licenses does not create a lawyer-client or
other relationship. Creative Commons makes its licenses and related
information available on an "as-is" basis. Creative Commons gives no
warranties regarding its licenses, any material licensed under their
terms and conditions, or any related information. Creative Commons
disclaims all liability for damages resulting from their use to the
fullest extent possible.

Using Creative Commons Public Licenses

=====

Creative Commons Attribution 4.0 International Public License

By exercising the Licensed Rights (defined below), You accept and agree
to be bound by the terms and conditions of this Creative Commons
Attribution 4.0 International Public License ("Public License"). To the
extent this Public License may be interpreted as a contract, You are
granted the Licensed Rights in consideration of Your acceptance of
these terms and conditions, and the Licensor grants You such rights in
consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.

Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рисунок 30: Отмена



```
LICENSE      [BM--]  8 L:[327+32 359/359] *(16464/16464b) <EOF>      [*][X]

  c. No term or condition of this Public License will be waived and no
  failure to comply consented to unless expressly agreed to by the
  licensor.

  d. Nothing in this Public License constitutes or may be interpreted
  as a limitation upon, or waiver of, any privileges and immunities
  that apply to the Licensor or You, including from the legal
  processes of any jurisdiction or authority.

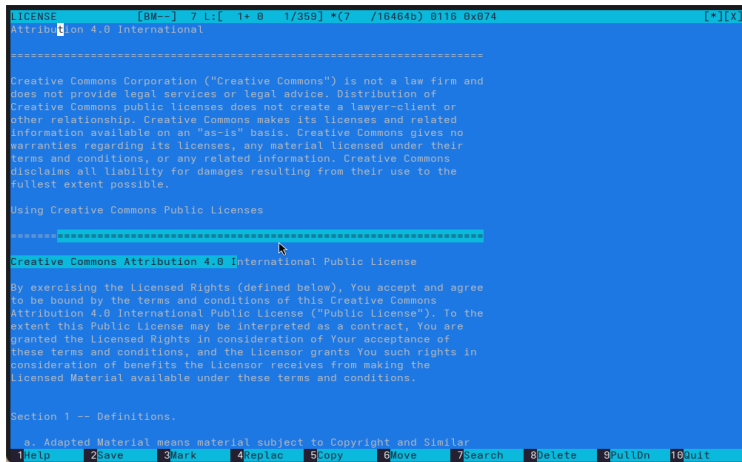
=====

Creative Commons is not a party to its public
licenses. Notwithstanding, Creative Commons may elect to apply one of
its public licenses to material it publishes and in those instances
will be considered the "Licensor." The text of the Creative Commons
public licenses is dedicated to the public domain under the CCB Public
Domain Dedication. Except for the limited purpose of indicating that
material is shared under a Creative Commons public license or as
otherwise permitted by the Creative Commons policies published at
creativecommons.org/policies, Creative Commons does not authorize the
use of the trademark "Creative Commons" or any other trademark or logo
of Creative Commons without its prior written consent including,
without limitation, in connection with any unauthorized modifications
to any of its public licenses or any other arrangements,
understandings, or agreements concerning use of licensed material. For
the avoidance of doubt, this paragraph does not form part of the
public licenses.

Creative Commons may be contacted at creativecommons.org.

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 31: Переход в конец файла



```
LICENSE      [BM--] 7 L:[ 1+ 0 1/359] *(7 /16464b) 0116 0x074      [*][X]
Attribution 4.0 International

=====

Creative Commons Corporation ("Creative Commons") is not a law firm and
does not provide legal services or legal advice. Distribution of
Creative Commons public licenses does not create a lawyer-client or
other relationship. Creative Commons makes its licenses and related
information available on an "as-is" basis. Creative Commons gives no
warranties regarding its licenses, any material licensed under their
terms and conditions, or any related information. Creative Commons
disclaims all liability for damages resulting from their use to the
fullest extent possible.

Using Creative Commons Public Licenses

=====
Creative Commons Attribution 4.0 International Public License

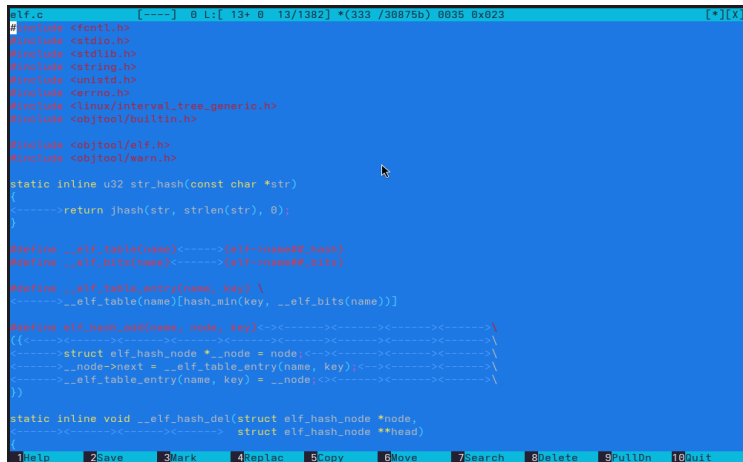
By exercising the Licensed Rights (defined below), You accept and agree
to be bound by the terms and conditions of this Creative Commons
Attribution 4.0 International Public License ("Public License"). To the
extent this Public License may be interpreted as a contract, You are
granted the Licensed Rights in consideration of Your acceptance of
these terms and conditions, and the Licensor grants You such rights in
consideration of benefits the Licensor receives from making the
Licensed Material available under these terms and conditions.

Section 1 -- Definitions.

a. Adapted Material means material subject to Copyright and Similar

1Help      2Save      3Mark      4Replac      5Copy      6Move      7Search      8Delete      9PullDn     10Quit
```

Рисунок 32: Переход в начало файла



```
elf.c [----] 0 L: [ 13+ 0 13/1382] *(333 /30875b) 0035 0x023 [*][X]
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <errno.h>
#include <linux/interval_tree_generic.h>
#include <objtool/builtin.h>

#include <objtool/elf.h>
#include <objtool/warn.h>

static inline u32 str_hash(const char *str)
{
    ----->return jhash(str, strlen(str), 0);
}

#define __elf_table(name)----->{elf->name##_hash}
#define __elf_bits(name)----->{elf->name##_bits}

#define __elf_table_entry(name, key) \
    ----->__elf_table(name)[hash_min(key, __elf_bits(name))]

#define elf_hash_add(name, node, key)----->----->----->\
    ({----->----->----->----->----->\
    ----->struct elf_hash_node *__node = node;----->----->----->\
    ----->__node->next = __elf_table_entry(name, key);----->----->----->\
    ----->__elf_table_entry(name, key) = __node;----->----->----->\
    })

static inline void __elf_hash_del(struct elf_hash_node *node,
    ----->----->-----> struct elf_hash_node **head)
```

1Help 2Save 3/ark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit

Рисунок 33: Файл с программой

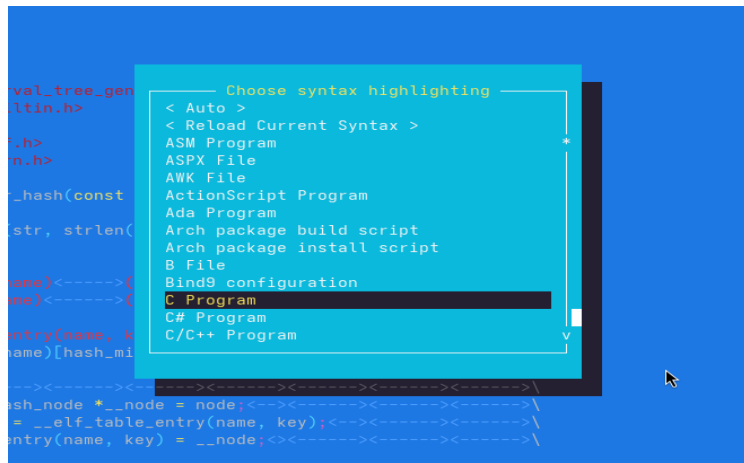


Рисунок 34: Цветовыделение синтаксиса

3. Выводы по проделанной работе

В данной работе мы ознакомились с инструментами поиска файлов и фильтрации текстовых данных. А также приобрели практические навыки по управлению процессами.